

LIBRARY MANAGEMENT SYSTEM

Python Project

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About the Project

This is a simple Library Management System made in Python. It is used to add, view, search, issue and return books.

The library has pre-added books for different engineering departments. The user can also add new books.

Simple Program Flow

START	Open the program
↓	Show menu
↓	Choose an option
↓	Do the selected work
↓	Show result / Process done
↓	Return to menu

Main Options

1. Add Book
2. View All Books
3. Search Book
4. Search By Department
5. Issue Book
6. Return Book
7. Exit

Program Code – Part 1

Book class and pre-added books

```
class Book:
    def __init__(self, name, author, department):
        self.name = name
        self.author = author
        self.department = department
        self.issued = False
        self.roll = ""
        self.days = 0

# Pre-added books, i searched all the books name and their author on google
books = [
    # Mechanical Engineering
    Book("Engineering Mechanics", "R.S. Khurmi", "Mechanical"),
    Book("Theory of Machines", "R.S. Khurmi", "Mechanical"),
    Book("Strength of Materials", "R.K. Rajput", "Mechanical"),
    Book("Fluid Mechanics", "R.K. Rajput", "Mechanical"),

    # Civil Engineering
    Book("Surveying", "B.C. Punmia", "Civil"),
    Book("Building Materials", "S.K. Duggal", "Civil"),
    Book("Structural Analysis", "R.C. Hibbeler", "Civil"),
    Book("Strength of Materials", "R.K. Bansal", "Civil"),

    # Electrical Engineering
    Book("Electrical Technology", "B.L. Theraja", "Electrical"),
    Book("Electrical Machinery", "P.S. Bimbhra", "Electrical"),
    Book("Engineering Circuit Analysis", "Hayt", "Electrical"),
    Book("Power System Engineering", "Nagrath and Kothari", "Electrical"),

    # Computer Science Engineering
    Book("Python Crash Course", "Eric Matthes", "Computer Science"),
    Book("Clean Code", "Robert C. Martin", "Computer Science"),
    Book("Operating System Concepts", "Silberschatz", "Computer Science"),
    Book("Computer Networking", "Kurose and Ross", "Computer Science"),
```

Program Code – Part 2

More pre-added books and Add / View functions

```
# Electronics Engineering
Book("Microelectronic Circuits", "Sedra and Smith", "Electronics"),
Book("Digital Design", "Morris Mano", "Electronics"),
Book("Electronic Devices", "Thomas Floyd", "Electronics"),
Book("Communication Systems", "Simon Haykin", "Electronics"),

# Chemical Engineering
Book("Chemical Engineering Thermodynamics",
     "Smith and Van Ness", "Chemical"),
Book("Unit Operations", "McCabe, Smith and Harriott", "Chemical"),
Book("Chemical Reaction Engineering",
     "Octave Levenspiel", "Chemical"),
Book("Transport Phenomena",
     "Bird, Stewart and Lightfoot", "Chemical"),

# Aerospace Engineering
Book("Introduction to Flight", "John D. Anderson", "Aerospace"),
Book("Fundamentals of Aerodynamics",
     "John D. Anderson", "Aerospace"),
Book("Aircraft Structures", "T.H.G. Megson", "Aerospace"),
Book("Flight Stability and Control",
     "Robert C. Nelson", "Aerospace"),

# Automobile Engineering
Book("Automotive Mechanics", "William H. Crouse", "Automobile"),
Book("Automobile Engineering", "Kirpal Singh", "Automobile"),
Book("Automotive Technology",
     "James D. Halderman", "Automobile"),
Book("Internal Combustion Engines",
     "V. Ganesan", "Automobile")
]

# this function adds a new book to the library
def add_book():
    name = input("Enter book name: ")
    author = input("Enter author name: ")
    department = input("Enter department: ")

    book = Book(name, author, department)
    books.append(book)

    print("\nBook added successfully!")
    print("Process done.")

# this function views all books in the library
def view_books():
    if len(books) == 0:
        print("No books in library.")
        return

    print("\n===== ALL BOOKS =====")

    for book in books:
        print("\nBook:", book.name)
        print("Author:", book.author)
        print("Department:", book.department)

        if book.issued == True:
            print("Status: Issued")
            print("Roll Number:", book.roll)
            print("Days:", book.days)
        else:
            print("Status: Available")

    print("-----")
```

Program Code – Part 3

Search, Issue, Return and Main Program

```
# this function searches for a book
def search_book():
    name = input("Enter book name: ")

    for book in books:
        if book.name.lower() == name.lower():
            print("\nBook found!")
            print("Book:", book.name)
            print("Author:", book.author)
            print("Department:", book.department)

            if book.issued == True:
                print("Status: Issued")
                print("Roll Number:", book.roll)
                print("Days:", book.days)
            else:
                print("Status: Available")

            print("Process done.")
            return

    print("Book not found.")

# this function searches books by department
def search_department():
    department = input("Enter department: ")
    found = False

    print("\n===== BOOKS =====")

    for book in books:
        if book.department.lower() == department.lower():
            print("\nBook:", book.name)
            print("Author:", book.author)
            print("Department:", book.department)

            if book.issued == True:
                print("Status: Issued")
            else:
                print("Status: Available")

            found = True

    if found == True:
        print("\nProcess done.")
    else:
        print("No books found for this department.")

# this function issues a book
def issue_book():
    name = input("Enter book name: ")

    for book in books:
        if book.name.lower() == name.lower():
            if book.issued == True:
                print("Book is already issued!")
                return

            book.roll = input("Enter student roll number: ")
            book.days = input("How many days do you want the book? ")
            book.issued = True

            print("\nBook issued successfully!")
            print("Book:", book.name)
            print("Student Roll Number:", book.roll)
            print("Number of Days:", book.days)
            print("Process done.")
            return

    print("Book not found.")

# this function returns a book
def return_book():
    name = input("Enter book name: ")

    for book in books:
        if book.name.lower() == name.lower():
```

```

        if book.issued == True:
            book.issued = False
            book.roll = ""
            book.days = 0

            print("\nBook returned successfully!")
            print("Process done.")
        else:
            print("This book is not issued.")
        return

    print("Book not found.")

# Main program
while True:
    print("\n=====")
    print("          LIBRARY MANAGEMENT SYSTEM")
    print("=====")

    print("1. Add Book")
    print("2. View All Books")
    print("3. Search Book")
    print("4. Search By Department")
    print("5. Issue Book")
    print("6. Return Book")
    print("7. Exit")

    choice = input("\nEnter your choice: ")

    if choice == "1":
        add_book()
    elif choice == "2":
        view_books()
    elif choice == "3":
        search_book()
    elif choice == "4":
        search_department()
    elif choice == "5":
        issue_book()
    elif choice == "6":
        return_book()
    elif choice == "7":
        print("\nThank you for using the Library Management System!")
        print("Program ended.")
        break
    else:
        print("Invalid choice! Please enter 1-7.")

```